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Revealing unseen dynamical regimes of ecosystems from population time-series data

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Many dynamical systems can exist in alternative regimes for which small changes in an environmental driver can cause sudden jumps between regimes. In ecology, predicting the regime of population fluctuations under unobserved levels of an environmental driver has remained an unsolved challenge with important implications for conservation and management. Here, we show that integrating time-series data and information on a putative driver into a Gaussian Process regression model for the system's dynamics allows us to predict dynamical regimes without the need to specify the equations of motion of the system. As a proof of concept, we demonstrate that we can accurately predict fixed-point, cyclic, or chaotic dynamics under unseen levels of a control parameter for a range of simulated population dynamics models. For a model with an abrupt population collapse, we show that our approach goes beyond an early warning signal by characterizing the regime that follows the tipping point. We then apply our approach to data from an experimental microbial food web and from a lake planktonic food web. We find that we can reconstruct transitions away from chaos in the microbial food web and anticipate the dynamics of the oligotrophic regime in the planktonic food web. These results lay the groundwork for making rational decisions about preventing, or preparing for, regime shifts in natural ecosystems and other dynamical systems.

regime shifts | population dynamics | empirical dynamic modeling | early warning signal | tipping points

Significance

Small changes in conditions can cause dynamical systems to suddenly shift to new, potentially undesirable, regimes. However, characterizing previously unseen dynamical regimes—a critical bit of information in evaluating the utility of costly interventions—remains an open challenge. Here, we focus on ecosystems and introduce a data-driven modeling approach that can extrapolate the effect of environmental conditions on population growth, allowing us to predict how populations will fluctuate under unobserved conditions. We test our approach with simulations and apply it to empirical data from microbial and planktonic food webs. By anticipating the dynamics that follows a tipping point (e.g., plankton fluctuations after a lake transition), we illustrate how our approach can facilitate decision-making in ecosystems with multiple regimes.

Dynamical systems in physics, climate, and ecology can exist in several distinct alternative regimes, and rapid shifts from one regime to another may result from gradual changes in some environmental driver or "control parameter" (1–3). In ecology, well-documented examples of regime shifts include clear macrophyte-dominated lakes transitioning into turbid phytoplankton-dominated lakes due to nutrient input (4), kelp forests becoming sea urchin barrens due to predator loss (5), coral reefs giving way to macroalgae reefs as a result of warming (6), and forests shifting to savannas after changes in precipitation (7). Because ecosystem services depend greatly on the state of the system, there has been considerable interest in understanding and predicting the dynamical regimes that may emerge under changing environmental conditions.

Predicting regime shifts in time to avert them has been an important avenue of research and many Early Warning Signals (EWS) have been developed to do so (1). These signals can be computed directly from time-series data and include variance, skewness, the AR(1) coefficient, and the power spectrum (8). The unifying idea behind EWS is that changes in the control parameter cause the system to cross a tipping point (i.e., a bifurcation) and most are based on "critical slowing down"—in which the stability of a fixed point relaxes as the bifurcation is approached (9). Empirical evidence for critical slowing down has been found in plankton in lakes (10), photoinhibition in cyanobacterial populations (11), declining food sources for *Daphnia magna* (12), and increased mortality in yeast (13). A key feature of EWS is that they are generic, that is, the dynamical behavior close to a bifurcation is well described by a handful of canonical "normal forms" (14) such that we do not need detailed knowledge of the underlying dynamics to anticipate a shift. However, EWS pertain to a relatively narrow range of bifurcations—namely the fold, transcritical, and Hopf bifurcations (3, 15)— and approaches that apply to a wider range of cases are currently needed. Most importantly, EWS only tell us that the system might be approaching a regime shift but give no information about what the upcoming regime will be.

Predicting the dynamics of ecosystems under unobserved values of an environmental driver, such as beyond a proposed tipping point, has been a major challenge in ecology. Although recent progress has been made in anticipating the composition and structure of certain ecosystems under unseen conditions (16, 17), taking the next step and predicting

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population fluctuations remains notoriously difficult. The difficulty stems from accurately extrapolating the effect of the driver on species abundances using data on few species and over a limited range of the driver. Unfortunately, most EWS are unable to perform this extrapolation (18, 19) and the few approaches that do [e.g., approaches leveraging deep learning (20)] require prior knowledge of the class of bifurcations. Although model-based approaches provide more specific predictions (3, 21), these require detailed knowledge about the underlying dynamics and are unlikely to be transferable across ecosystems. Therefore, an alternative approach, with generality similar to that of existing EWS, but that also predicts the dynamics beyond a tipping point is highly desirable.

Empirical Dynamic Modeling (EDM) is a framework based on nonparametric function approximation and time-delay embedding that has proved useful to understand and forecast a wide range of dynamical systems. Examples include forecasting recruitment in fisheries (22), understanding the environmental drivers of diseases (23), inferring causal interaction networks (24), and quantifying population responses to perturbations (25). By learning about system dynamics directly from time-series data (through nonparametric function approximation) and by accounting for unobserved state variables (through time-delay embedding), EDM can give insights about the dynamics without the need to specify a model, as with EWS. Several studies have used EDM to develop new EWS, which include approaches based on prediction accuracy (26), nonlinearity (27), and eigenvalues of the Jacobian matrix (28). Other recent studies have used EDM to detect whether and when a regime shift has happened in the past based on prediction accuracy tests (29, 30). What is currently lacking from both EWS and EDM approaches is the capacity to predict dynamics at novel driver levels. Yet, being able to leverage information on putative drivers (e.g., temperature, nutrients, rainfall, fishing pressure) could allow us to anticipate unseen regimes and at the same time advance our understanding about a given system.

Here we introduce an approach based on EDM that allows us to predict the dynamical behavior of an ecosystem at unobserved levels of an environmental driver. Our approach works by first integrating population time-series data and information on the putative driver into a Gaussian Process regression with time-delay embedding [GP-EDM, (31)] to learn how the driver affects population dynamics. We then predict the dynamics at unobserved levels of the driver by interpolating or extrapolating the GP-EDM model. Using several population dynamics models, we demonstrate that we can accurately predict unseen fixed-point, cyclic, or chaotic regimes, in addition to predicting the location of multiple tipping points in a bifurcation diagram. We also show that, for a model with an abrupt population collapse, our approach can anticipate the regime that follows the tipping point. Unsurprisingly, the accuracy of these predictions relies on informative training data and declines with long-range extrapolation. Finally, we apply our approach to empirical time series from both an experimental microbial and a lake planktonic food web. We show that we can reconstruct transitions away from chaos in the experimental food web and anticipate the dynamics of the oligotrophic regime in the lake food web.

Results

Testing the Approach on Model-Generated Data. We first tested the ability of our approach to predict unseen dynamical regimes and the location of tipping points using four discrete-time population dynamics models (*SI Appendix*, section S1). Here, we focus on a two-species predator–prey model and detailed results for other models can be found in *SI Appendix*. For each model, we trained a GP-EDM model [*SI Appendix*, section S3; (31)] using time series from a single species (x_i) generated at four different levels of an environmental driver (i.e., a control parameter p; Fig. 1*A* and *SI Appendix*, Fig. S1). The trained GP-EDM models produced highly accurate leave-one-out predictions under the observed data (R^2: single-species model, 0.992; two-species competition model with harvesting, 0.998; two-species predator–prey model, 0.979; three-species competition model, 0.954; *SI Appendix*, Fig. S2). Note that we use R^2 here to indicate the mean squared error of leave-one-out predictions divided by the total time-series variance (Eq. **4**) and that model selection based on out-of-sample predictions is standard in EDM (32–34). The chosen embedding dimension ($E + 1$) for each model is shown in *SI Appendix*, Fig. S2 and plots of model residuals in *SI Appendix*, Fig. S3. Then, using the delay-embedding map G_i (Eq. **3**) approximated via GP-EDM (Fig. 1*B*), we performed predictions of x_i over unobserved values of p to reconstruct the bifurcation diagram (Fig. 1 *C* and *D* and *SI Appendix*, Fig. S1). Although G_i clearly has a different shape than the true map (F_i, Eq. **2**), Fig. 1*B* shows that these functions change in analogous ways as p is increased. In *SI Appendix*, Fig. S1, we illustrate the case where G_i and F_i have the same inputs and, therefore, are more directly comparable to each other.

We were able to accurately predict the dynamics of x_i across many unobserved levels of p for all four models (Fig. 2 and *SI Appendix*, Fig. S4). For instance, having observed the prey time series at four regimes of the two-species predator–prey model (fixed points at $p = 2.4$ and $p = 2.6$, cycle at $p = 3.15$, and chaos at $p = 3.4$), we accurately predicted unobserved fixed-point, cyclic, or chaotic regimes (Fig. 2*A*). We assessed the reconstructed bifurcation diagram using the Jensen–Shannon divergence (JSD) and found that the distribution of predicted abundances of species i ($\{\hat{x}_i(t)\}$, $t = 1, ..., T$) for each p significantly matched the distribution of true abundances ($\{x_i(t)\}$, $t = 1, ..., T$; Fig. 2*B* and *SI Appendix*, Fig. S4). Note that an accurate reconstruction of the dynamics of species i does not imply an accurate reconstruction of the dynamics of other unobserved species. Nevertheless, we obtained similar results when changing the species used to perform time-delay embedding (*SI Appendix*, Fig. S5). As expected, our approach was most accurate when we used all species in the GP-EDM model (i.e., native coordinates; *SI Appendix*, Fig. S6) instead of using only a single species with time-delay embedding (i.e., delay coordinates; *SI Appendix*, Fig. S4). In addition to low JSD, predicted regimes showed dynamics with a similar time scale (*SI Appendix*, Fig. S7), periodicity (*SI Appendix*, Fig. S8), and stability (*SI Appendix*, section S4 and Fig. S8) as the true regimes. Although with slightly less accuracy, we also detected the location of tipping points (i.e., bifurcations) that separate different regimes. This includes the location of a Hopf bifurcation (Fig. 2), period doubling bifurcations (*SI Appendix*, Fig. S4 *A* and *D*), and a fold bifurcation associated with population collapse (*SI Appendix*, Fig. S4*B*).

We evaluated the sensitivity of our results by performing the above analysis for different dynamical regimes in our training data (*SI Appendix*, Fig. S9). We found that we can most accurately predict unseen regimes and detect the location of tipping points when the training data includes a chaotic regime, which provides information for the GP-EDM model across the entire range of x_i. We also performed these analyses for several different training datasets with different amounts of process noise. Although high noise can make our approach less precise, we found that, on average, the JSD between true and predicted

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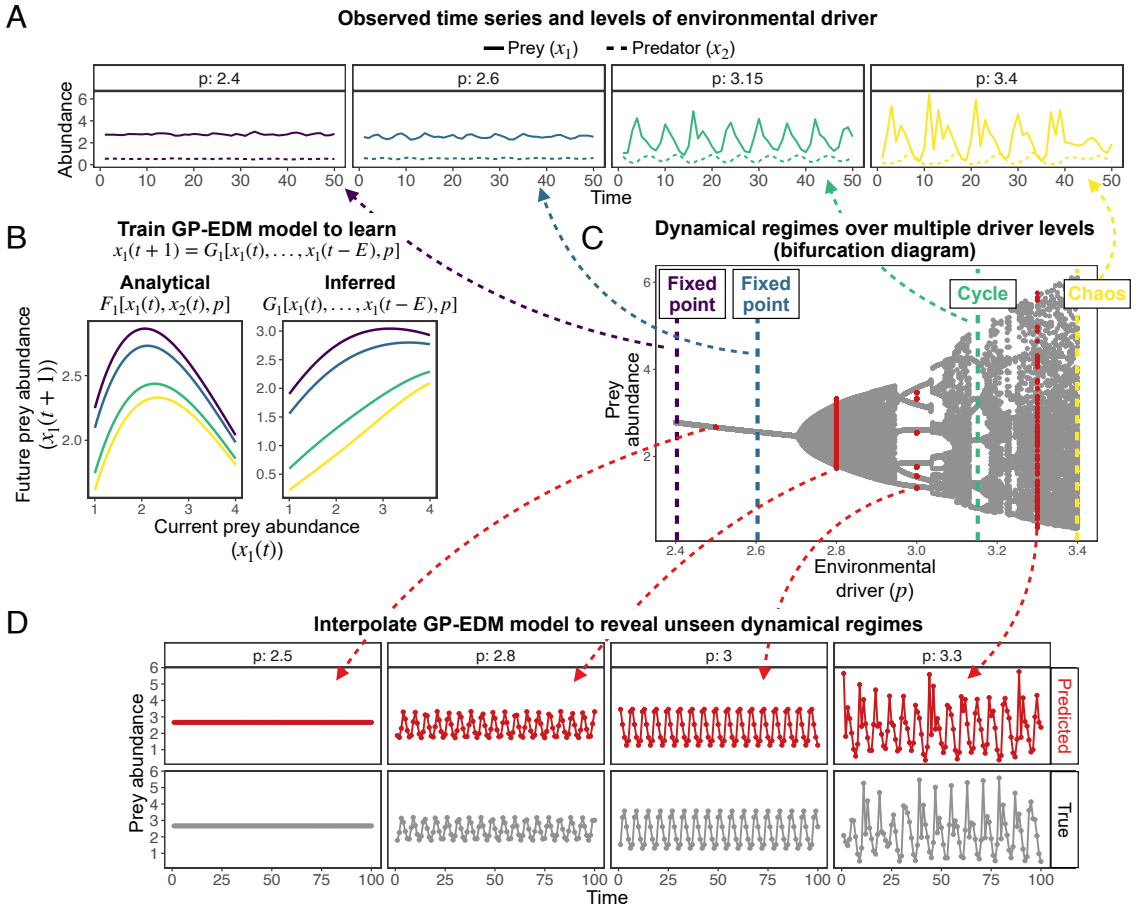

Fig. 1. GP-EDM approach to reveal unseen dynamical regimes. (*A*) Population time series generated from a predator–prey model (*SI Appendix*, section S1) at four different levels of the attack rate parameter (i.e., the environmental driver p). Low values of p generate fixed points ($p = 2.4$ in purple and $p = 2.6$ in blue) whereas high values of p generate cycles ($p = 3.15$ in green) or chaos ($p = 3.4$ in yellow). (*B*) *Left* panel depicts the true relationship between current ($x_1(t)$) and future ($x_1(t+1)$) prey abundance at different levels of p (different colors). *Right* panel depicts the same relationship but obtained from the trained GP-EDM model. Note that $x_2(t)$ is also an input of function F_1 and $x_1(t-1)$, $x_1(t-2)$, and $x_1(t-3)$ are also inputs of function G_1, but these inputs are held constant here for simplicity. (*C*) Bifurcation diagram of the predator–prey model showing the different dynamical regimes that emerge as we vary p (gray points). The four regimes shown in (*A*) are represented as vertical dashed lines. We can then interpolate the trained GP-EDM model to unseen levels of p, which are shown in red. (*D*), Predicted dynamical regimes at unseen levels of p (in red) match closely the true dynamics (in gray). These regimes are fixed point ($p = 2.5$), low-amplitude cycle ($p = 2.8$), high-amplitude cycle ($p = 3.0$), and chaos ($p = 3.3$). Note that the correct comparison is between the red and gray dynamical regimes and not exact trajectories, given that initial conditions are arbitrary.

dynamics was low (*SI Appendix*, Fig. S10). Finally, we obtained an accurate reconstruction of the bifurcation diagram when using a continuous-time model (*SI Appendix*, section S2 and Fig. S11).

We then tested whether our approach can be used as an EWS of an upcoming population collapse using a two-species competition model with harvesting (*SI Appendix*, section S1). To do so, we generated a time series for which the harvest rate (i.e., the control parameter p) changed linearly from 0.42 to 0.48 and a tipping point occurred at $p = 0.464$ according to a change-point analysis (Fig. 3*A*). We then sequentially updated our GP-EDM model as the harvest rate increased toward the tipping point and, for each update, we extrapolated the entire bifurcation diagram (i.e., all dynamical regimes from $p = 0.42$ to $p = 0.48$). Again, the best trained GP-EDM model produced highly accurate leave-one-out predictions ($R^2 = 0.986$).

Our approach was able to anticipate population collapse when the training data ended 4 time steps before the tipping point (Fig. 3*B*). Although the predicted abundance at the unseen regime was higher than the true abundance, our approach clearly anticipated an abrupt abundance shift. The harvest rate of population collapse was predicted to be $p = 0.4608$, which is slightly below the tipping point location (i.e., $p = 0.464$). When the training data ended exactly at the tipping point, our approach predicted a more accurate tipping point location at $p = 0.465$. Finally, when the training data contained the collapsed regime, our approach provided a very accurate reconstruction of the entire bifurcation diagram, as expected. Note that the tipping point location ($p = 0.464$) will be different for every simulation due to process noise. We obtained similar although slightly less accurate results when using training data with tipping points at different locations (*SI Appendix*, Fig. S12). Finally, the rolling variance of abundance only started to increase after the tipping point (Fig. 3*A*), suggesting a limited use of this EWS under this particular model.

Applying the Approach to an Experimental Microbial Ecosystem. Next, we applied our approach to population time-series data from a seminal chemostat experiment that discovered transitions from fixed point to chaos, then to cycles (p; i.e., mortality) was reduced (35). This dataset is analogous to our simulation set up from Fig. 2 and, therefore, we used the same analytical approach (*Materials and Methods*).

For each of the three species in this experimental food web, we obtained leave-one-out predictions that are 49 to 67% more

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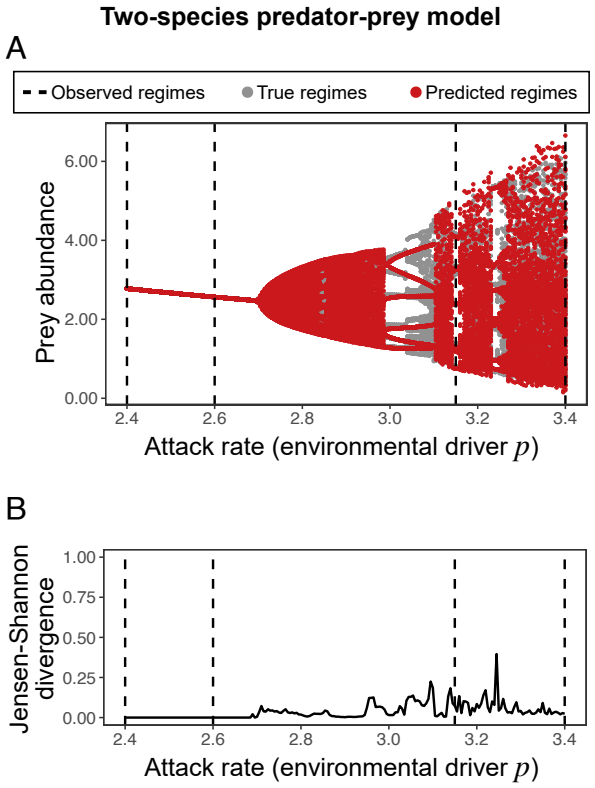

Fig. 2. GP-EDM approach accurately predicts a range of unseen dynamical regimes. (*A*) True (in gray) and predicted (in red) bifurcation diagram for a two-species predator–prey population dynamics model (*SI Appendix*, section S1). The bifurcation diagram depicts the abundance of the prey species after the transient period (y-axis) for a range of control parameter values (p, x-axis). (*B*) Jensen–Shannon divergence (JSD) computed at each value of p. The JSD ranges from 0 to 1, where 0 represents a close match between true and predicted dynamics. The average JSD across all values of p is 0.037. In (*A* and *B*) the vertical dashed lines depict the values of p for which we observed population time series to train the GP-EDM model (as in Fig. 1*C*).

accurate than simply using their average abundance as the prediction (R^2: predator, 0.490; preferred prey, 0.508; less-preferred prey, 0.670; *SI Appendix*, Fig. S13). The chosen embedding dimension ($E + 1$) for each species is shown in *SI Appendix*, Fig. S13 and plots of model residuals in *SI Appendix*, Fig. S14. This suggests that GP-EDM is reasonably approximating the function G_i (Eq. **3**) at the observed dilution rates. More importantly, the bifurcation diagrams reconstructed independently from each species exhibit a large region of chaos around $p = 0.5$/day, cycles around $p = 0.45$/day and slightly above $p = 0.6$/day, and a fixed point for $p > 0.7$/day (Fig. 4). Our results also reveal the dilution rate levels at which we should observe unseen bifurcations. For example, the appearance of a fixed point is predicted to occur somewhere between $p = 0.62$/day and $p = 0.68$/day (Fig. 4). Although we cannot verify this experimentally, the fact that similar results were obtained independently for each species increases our confidence that the approach accurately illuminates previously invisible regimes and bifurcations.

We evaluated whether these results were robust to how we scaled species abundances and assigned prior information to p as a model input. Results for predator and preferred prey species were robust to these changes, whereas results for the less-preferred prey species were not (*SI Appendix*, Fig. S15). We also excluded a given dilution rate from our training data and repeated our analyses to verify whether we could predict the regime under the excluded dilution rate. We could accurately predict the fixed-point and cyclic regimes of all species (i.e., $p = 0.45$, $p = 0.75$, and $p = 0.9$), but not the chaotic regime (i.e., $p = 0.5$; *SI Appendix*, Fig. S16). Hence, similar to our model results, we found that regimes that explore the entire state space (whether through chaos or process noise) are the most informative pieces of the training data.

Applying the Approach to a Lake Planktonic Ecosystem. For our last set of analyses, we applied the EWS version of our approach to the well-known case of lakes that shift from an eutrophic to an oligotrophic regime. We used monthly time-series data from Lake Zurich of 13 plankton functional groups, phosphate concentration (i.e., the control parameter p), and water temperature from January, 1978 to December, 2019 [*SI Appendix*, Fig. S17; $T = 504$ data points; (16)]. Because this planktonic food web is thought to have shifted from an eutrophic to an oligotrophic regime via a gradual reduction of phosphate concentration (16), it is analogous to our simulations from Fig. 3.

We first performed a change-point analysis to determine the location of a potential tipping point for each functional group. We found that large green algae had the most well-resolved shift followed by small cryptophytes and omnivores (*SI Appendix*, Table S1). This potential tipping point for large green algae occurred in November, 1987 ($t_{split} = 119$) at a phosphate concentration of 0.0472 mg/l (Fig. 5*A*). The potential tipping points for small cryptophytes (October, 1988; $t_{split} = 130$) and omnivores (June, 1986; $t_{split} = 102$) occurred around the same time, suggesting an ecosystem-wide shift.

To separate the effects of the control parameter from seasonality, we first applied a centered moving average smoother to the phosphate time series, removing seasonal oscillations while retaining the trend. We included seasonality in our GP-EDM model by adding water temperature as an input. We then used data from January, 1978 to October, 1986 (106 data points) to train GP-EDM models and verify which functional groups were most predictable. Small green algae showed the best ($R^2 = 0.801$), whereas large green algae showed the second-best ($R^2 = 0.770$; *SI Appendix*, Fig. S18) leave-one-out prediction accuracy (*SI Appendix*, Table S2). Due to strong evidence of a tipping point for large green algae from the change-point analysis, we focused subsequent analyses on this functional group. The chosen embedding dimension and plots of model residuals for large green algae are shown in *SI Appendix*, Table S2 and Fig. S18. The GP-EDM model suggested that higher phosphate concentrations increase the nonlinearity in density dependence of large green algae (*SI Appendix*, Fig. S19).

We then reconstructed bifurcation diagrams for large green algae using successively larger training datasets (Fig. 5*A*). We found evidence of a previously unseen low abundance regime when the training data ended 1 y before the potential tipping point (i.e., training data up to October, 1986; Fig. 5*B*). We then clearly predicted an abrupt shift in the dynamics of large green algae when the training data ended 1 mo before the potential tipping point (i.e., training data up to October, 1987; Fig. 5*B*). This abrupt shift was predicted to occur at a phosphate concentration of 0.0470 mg/l, very close to the phosphate concentration at the potential tipping point detected via change-point analysis. Interestingly, the bifurcation diagram reconstructed with training data up to October, 1987 was very similar to the bifurcation diagram reconstructed with

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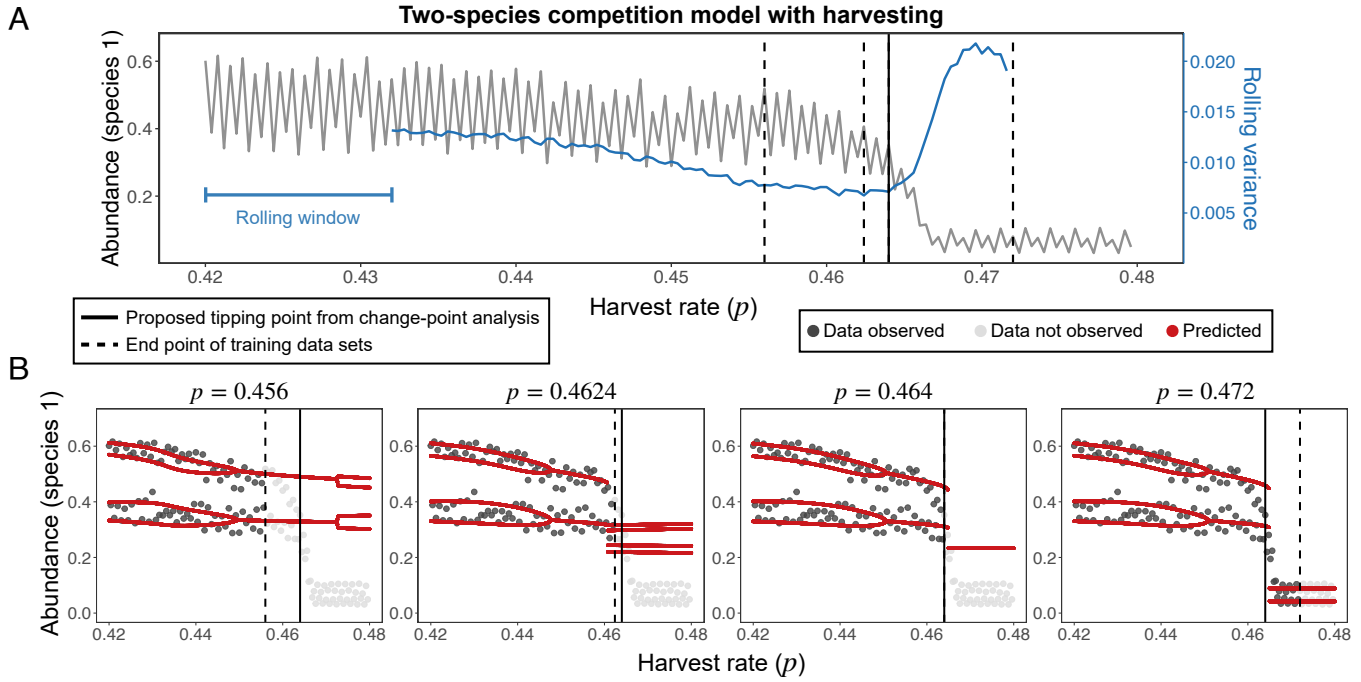

Fig. 3. Illustration of GP-EDM approach as an Early Warning Signal. (*A*) Population time series (in gray) of the harvested species in a two-species competition model (*SI Appendix*, section S1). Harvest rate (p) is depicted in the x-axis and is proportional to time. The population shows a tipping point at $p = 0.464$ (vertical solid line) according to a change-point analysis. The vertical dashed lines show the end points of the training datasets used with the GP-EDM model. The rolling variance (a typical EWS) computed using a window with 30 points is shown in blue. (*B*) Reconstructed bifurcation diagrams (in red) using the GP-EDM model trained up to each of the four dashed lines. Bifurcation diagrams depict the population abundance after the transient period (y-axis) for a range of harvest rate values (x-axis). The value of p at the end of each training dataset is shown above each plot. Gray points denote population abundance values used to train the GP-EDM model (dark gray) or not yet observed (light gray).

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training data up to October, 1996 (i.e., 9 y after the tipping point; Fig. 5*B*), reinforcing that the former predictions were accurate. In contrast to these results, the rolling variance of large green algae abundance—the EWS that we used in Fig. 3—did not steadily increase prior to the tipping point (*SI Appendix*, Fig. S20).

Our results were qualitatively the same when removing temperature as a model input, assigning prior information to p, and changing the phosphate moving average smoother (*SI Appendix*, Fig. S21). Last, we mimicked our analysis with the two-species predator–prey model (*SI Appendix*, Fig. S8) and computed the power spectrum and largest Lyapunov exponent of the predicted regimes of large green algae. We found evidence that the dynamics of large green algae transitions from chaos to stable cycles as phosphate concentration decreases during lake oligotrophication (*SI Appendix*, Fig. S22).

Discussion

Foreseeing the dynamical regimes of ecosystems under unobserved environmental conditions is an important ecological challenge, with applications in both conservation and management (2, 19, 36, 37). Although a large body of research has developed EWS that foretell an impending regime shift (1, 8), anticipating what type of dynamics we will observe after the shift has remained elusive. The GP-EDM approach introduced here integrates time-series data and information on a putative driver to enable interpolation and extrapolation to previously unseen dynamical regimes and serve as an EWS. In both simulated and empirical data, we found that GP-EDM can provide useful inference on when and how these systems change to a different regime.

Many EWS have been developed and employed to detect upcoming regime shifts, which include variance, skewness, the AR(1) coefficient, and the power spectrum (8). Although these indicators cannot give information about unseen regimes, other recent EWS based on deep learning or reservoir computing can perform this task (20, 38, 39). However, these approaches require large amounts of training data, information on the nonlinearities present in the underlying dynamics, or knowledge of the class of bifurcations. In contrast, GP-EDM can be applied with more modest amounts of data and is completely agnostic about the underlying dynamics. Therefore, our approach is able to identify bifurcations that are not among those typically considered in ecological studies. Most importantly, our approach leverages information on the putative driver of regime shifts and, by doing so, allows us to better understand the range of conditions under which an ecosystem exhibits a given type of dynamics.

Although our results are promising, several limitations of GP-EDM are important to note. First, to be used as an EWS, the GP-EDM model must extrapolate to new regions of the control parameter (p) space. Since the prediction accuracy of any Gaussian Process regression declines with long-range extrapolation (40, 41), this limits our ability to anticipate distant regime changes. However, when the inverse-length scale parameter ϕ_i associated with p is small we can extrapolate much farther than when ϕ_i is large (*SI Appendix*, section S3). A small ϕ_i indicates that the map given by function F_i (Eq. **2**) is more linear in p. Second, we found that when the training data includes a limited range of abundances (e.g., when it contains mostly fixed points), GP-EDM is likely to determine that the control parameter has little to no effect and the resulting bifurcation diagram is inaccurate (*SI Appendix*, Fig. S9). Therefore, our approach is more accurate when the training data spans a large portion of the state space, either because of process noise or large deterministic fluctuations (e.g., chaos). Note that this is

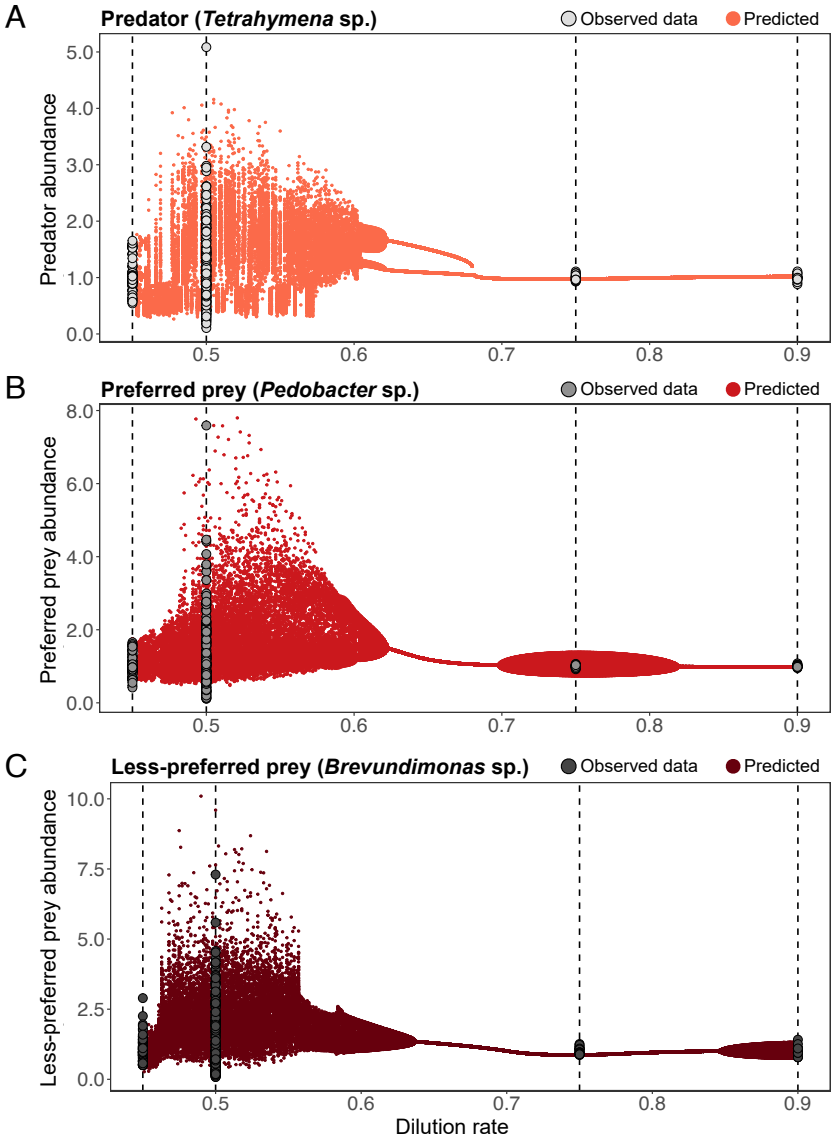

Fig. 4. Predicting unseen dynamical regimes in an experimental microbial food web. Bifurcation diagrams reconstructed with our GP-EDM approach (red points) for the predator species [*Tetrahymena* sp., light red, (*A*)], preferred prey species [*Pedobacter* sp., moderate red, (*B*)], and less-preferred prey species [*Brevundimonas* sp., dark red, (*C*)]. Each bifurcation diagram depicts the abundance of a given species after the transient period (y-axis) for a range of dilution rate values (x-axis). Vertical dashed lines represent the dilution rates of the experimental treatments (0.45/day, 0.5/day, 0.75/day, and 0.9/day). Gray points show the abundance values at these dilution rates used to train a GP-EDM model separately for each species. Abundances are in arbitrary units due to our data scaling process (*Materials and Methods*).

a typical requirement of statistical methods that aim to infer dynamical properties of a system from time-series data (38, 42). In situations where we know that the control parameter is important but GP-EDM determines that it has no effect, we suggest fixing ϕ_i associated with p to 1 (*SI Appendix*, section S3). A related point concerns the required time-series length. Based on previous EDM studies, we recommend using time series that covers several multiples of the organism's generation time for each level of the control parameter (22, 34). Third, ecosystems are typically under the effect of multiple environmental drivers. Our study, like most work in EWS, was restricted to a single control parameter that leads to codimension-one bifurcations. Future studies could include additional control parameters in the GP-EDM model to evaluate whether it is possible to anticipate more complicated regime shifts. Finally, in contrast to other EWS, GP-EDM requires data on the putative control parameter, or a suitable proxy. In the absence of a known driver, slow feature analysis (43) may be useful to construct a suitable proxy and future studies could explore this idea.

In addition to testing our approach with model-generated data (Figs. 2 and 3), we applied our approach to two empirical datasets: an experimental microbial ecosystem [Fig. 4; (35)] and a lake planktonic ecosystem [Fig. 5; (16)]. Both applications revealed alternative dynamical regimes and shifts between regimes in these ecosystems. Regarding the experimental microbial ecosystem, previous work had determined the existence of three qualitative distinct regimes [limit cycle at low dilution rate, chaos at intermediate dilution rate, and fixed point at high dilution rate; (35)]. However, no experiment was conducted to verify the exact location of bifurcations. Our analyses shed light on this question, especially regarding the shift from the chaotic to the fixed point regime, which should occur at a dilution rate of 0.62 to 0.68/day

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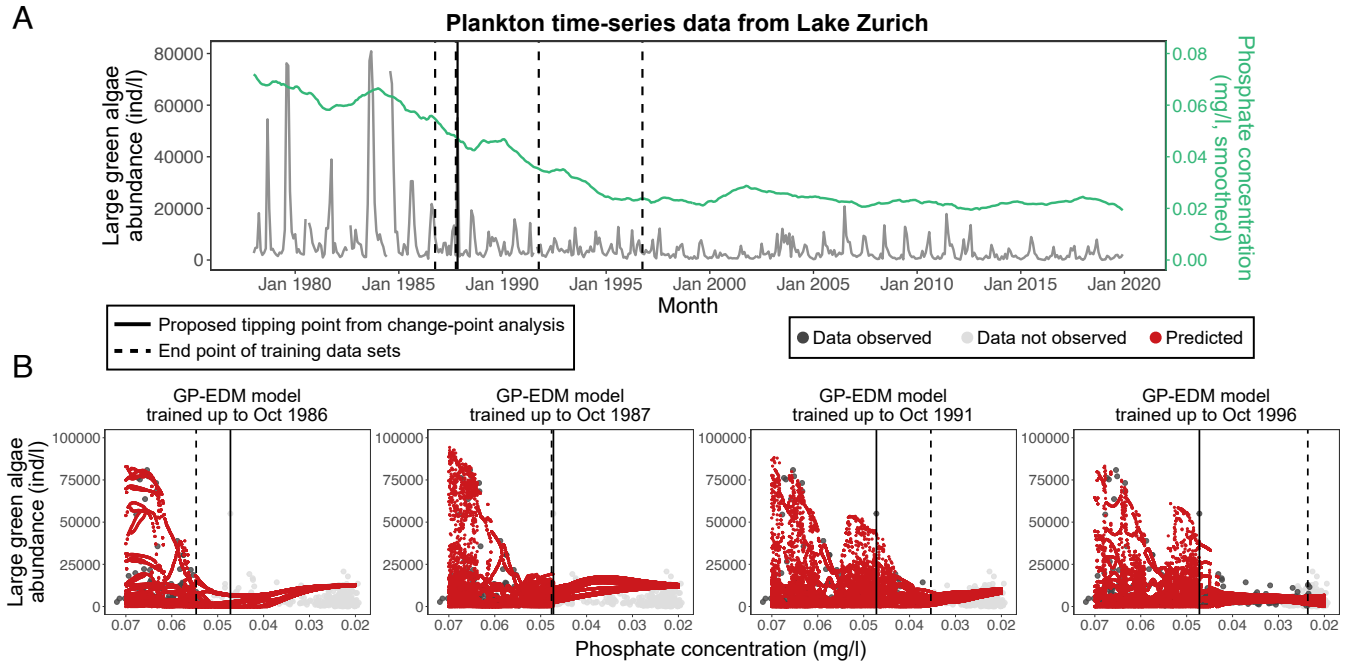

Fig. 5. Anticipating the oligotrophic regime in a lake planktonic food web. (*A*) Monthly population time series (in gray) of large green algae in Lake Zurich from January, 1978 to December, 2019. The population shows a tipping point on November, 1987 (vertical solid line) according to a change-point analysis. The vertical dashed lines show the end points of the training datasets used with the GP-EDM model. Phosphate concentration (smoothed via a centered moving average), a key environmental driver for oligotrophication, is shown in green. (*B*) Reconstructed bifurcation diagrams (in red) using the GP-EDM model trained up to each of the four dashed lines. Bifurcation diagrams depict the population abundance after the transient period (y-axis) for a range of phosphate concentration values (x-axis). The training data end points are shown as dates above each plot. Gray points denote population abundance values used to train the GP-EDM model (dark gray) or not yet observed (light gray).

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(Fig. 4). Future work could test our prediction by performing experiments under these dilution rates.

Regarding the lake planktonic ecosystem, a large body of research has explored the transitions from oligotrophic to eutrophic and vice versa (4, 16, 44, 45). When analyzing a complex ecosystem such as a lake food web with multiple functional groups, it is challenging to determine whether a shift has occurred, because only certain groups may show a clear pattern of a tipping point (19, 46–48). A key component of GP-EDM is that it uses time-delay embedding to account for hidden state variables. By focusing on large green algae—a highly predictable functional group that showed clear evidence of a tipping point—we were able to anticipate a possible ecosystem-wide regime shift. That is, one month before the potential tipping point, our approach clearly indicated an abrupt shift from a chaotic regime with high-amplitude abundance fluctuations to a cyclic regime with low-amplitude abundance fluctuations for large green algae. Note that this type of regime shift is common in phytoplankton under nutrient reductions but is hard to anticipate with traditional EWS such as rolling variance (*SI Appendix*, Fig. S20) (46, 48). We found that the shift from eutrophic to oligotrophic occurred at a phosphate concentration of 0.0470 mg/l. An important area for future research is to apply GP-EDM to a range of natural ecosystems for which there are also data on a known environmental driver that induces regime shifts to obtain cross-system insights. This type of study could, for example, elucidate whether there is a typical phosphate concentration for the eutrophic-oligotrophic transition in lakes.

In summary, we show that GP-EDM accurately reveals previously unseen dynamical regimes under changing environmental conditions across a range of complexity. Although regime shifts can be extremely costly, they may also be beneficial or require no intervention [e.g., eutrophic-oligotrophic transition in lakes (16) and fish recruitment success due to hydroclimate regime shift (49)]. Hence, in order to engage in appropriate mitigation actions, it is important to have a plausible estimate of the cost of inaction, which in turn requires us to predict the state of the ecosystem following an impending transition. In addition to ecosystems, many systems in physics, climate, and finance exhibit tipping points (1) and such cost–benefit analyses are useful in these cases as well. The GP-EDM approach developed here applies to any dynamical system modeled as Eq. **1** and future work could use this approach to anticipate unseen regimes in systems from other disciplines. By making accurate predictions of unobserved dynamical regimes, our approach opens the door to rational decision-making and cost–benefit analyses of regime shifts.

Materials and Methods

Dynamical Regimes and Bifurcations. Our goal in this study was to predict population dynamics that occurs under unobserved levels of an environmental driver. By doing so, we also provided predictions for the driver levels under which we should observe tipping points. To formalize these ideas, consider a generic discrete-time model that describes the population dynamics of n species in an ecosystem:

$$\mathbf{x}(t+1) = \mathbf{F}[\mathbf{x}(t), p], \qquad [1]$$

where $\mathbf{x}(t) = [x_1(t), ..., x_n(t)]^\top$ is the vector of abundances at time t, $\mathbf{F} = [F_1, ..., F_n]^\top$ is a vector-valued function, and p is a *control parameter*. Note that $\mathbf{F}$ typically depends on other parameters, which we consider to be fixed and omit here for simplicity. Systems modeled as Eq. **1** can exhibit different types of

attractor, including fixed points, cycles, and chaos. Each of these types of attractor may have distinct properties such as high or low abundance for fixed points and long or short periods for cycles. We define a *dynamical regime* as an attractor with a given qualitative property. A *bifurcation* occurs when a small change in the control parameter p in Eq. **1** causes a change in its dynamical regime–formally, when this parameter change results in a topologically different phase portrait (14). Specifically, a bifurcation occurs at a critical value (p_c) if we observe distinct dynamical regimes for $p_c - \delta$ and $p_c + \delta$, where δ is a small displacement. Note that we use the terms bifurcation and tipping point interchangeably throughout the text. The number of control parameters that must change to cause a bifurcation is known as the *codimension* of a bifurcation. When working with empirical data, we can leverage knowledge on an environmental driver (e.g., temperature, nutrients, rainfall, fishing pressure) that is known to cause regime shifts and assume that this driver is a proxy for p.

Here, we address the following challenge: Having observed time-series data $\{x_i(t)\}$ $(t = 1, ..., T)$ for a given species i at different dynamical regimes (i.e., at different values of p), can we predict regimes at unobserved values of p (Fig. 1)? We introduce an approach to solve this challenge by focusing on bifurcations of codimension one. In the next section, we describe our approach based on EDM to predict unseen dynamical regimes from time-series data.

Approach to Predict Unseen Dynamical Regimes. To address the problem described above, we combined Gaussian Process regression with time-delay embedding, an approach known as GP-EDM (31). Having observed time series $\{x_i(t)\}$ $(t = 1, ..., T)$ for a given species i at different values of p (Fig. 1*A*), we would like to learn the map F_i to be able to predict the dynamics of x_i at unobserved values of p:

$$x_i(t+1) = F_i[x_1(t), ..., x_n(t), p]. \quad \textbf{[2]}$$

However, if we only have data on species i, we propose that we can use time-delay embedding to compensate for unobserved species and instead learn the delay-embedding map G_i (Fig. 1*B*):

$$x_i(t+1) = G_i[x_i(t), ..., x_i(t-E), p], \quad \textbf{[3]}$$

where $E + 1$ is the number of lagged versions of x_i, also known as the *embedding dimension*. We can then apply an appropriate nonparametric function approximation method, such as Gaussian Process regression, to learn G_i and interpolate or extrapolate this function to unseen values of p (Fig. 1 *C* and *D*). We provide a detailed description of GP-EDM in *SI Appendix*, section S3, and we use the R package "GPEDM" to perform our analyses (50).

The rationale for this approach is as follows. Takens' theorem proves a one-to-one correspondence between an attractor in native coordinates (i.e., $x_1(t), ..., x_n(t)$) and one that has been reconstructed using time-delay embedding (i.e., $x_i(t), ..., x_i(t-E)$) (51). The bundle embedding theorem (52) extends time-delay embedding to forced dynamical systems and justifies the inclusion of the control parameter p in the reconstruction. Hence, a bifurcation in native coordinates corresponds to a bifurcation in delay coordinates at the same value of p. Therefore, we hypothesize that learning G_i and using it to interpolate between and extrapolate beyond observed values of p will allow us to reconstruct the bifurcation diagram and to predict the behavior of an ecosystem following a regime shift. Although a bifurcation corresponds to an abrupt change of the attractor, we expect the function G_i to change smoothly with p (Fig. 1*B*). Note that if we had data on all species in an ecosystem, then we could use Gaussian Process regression (without time-delay embedding) to learn F_i directly (instead of G_i). Also note that, although we use a simple description of our system as having only species abundances as state variables, Takens' theorem also works when the system contains other unobserved state variables (e.g., life stages, traits). Finally, since Takens' theorem also applies to continuous-time dynamics (51), our approach does as well. That is, we can use a discretely sampled series $\{x_i(t)\}$ $(t = 1, ..., T)$ to approximate the map that results from integrating the continuous-time dynamics with GP-EDM (34).

Testing the Approach on Model-Generated Data. We tested our approach on four discrete-time and one continuous-time population models exhibiting different types of bifurcations. Discrete-time models included a single-species model, a two-species competition model with harvesting, a two-species predator–prey model, and a three-species competition model (*SI Appendix*, section S1). Analyses for the continuous-time model are slightly different and are described in *SI Appendix*, section S2. For each discrete-time model, we first determined a range of parameter values, $[p_{min}, p_{max}]$, that produced different dynamical regimes. Then, we selected four distinct values of p (p_1, p_2, p_3, and p_4) within $[p_{min}, p_{max}]$ and simulated the dynamics with process noise to generate four time series of length $T = 50$. Note that we iterated each model for 500 time steps and discarded the first 450 points to remove transients. We then stacked these time series together along with additional inputs (i.e., lags of x_i and values of p) to create a data matrix to train the GP-EDM model (*SI Appendix*, Fig. S1). We standardized all inputs to zero mean and unit SD prior to fitting. We selected the value of E for the GP-EDM model based on leave-one-out prediction accuracy, measured as

$$R^2_E = 1 - \frac{\sum_{t=1}^{T}[x_i(t) - \hat{x}_i(t)]^2}{\sum_{t=1}^{T}[x_i(t) - \bar{x}_i]^2}, \quad \textbf{[4]}$$

where $\hat{x}_i(t)$ is the GP-EDM prediction for the left-out observation $x_i(t)$ using a given value of E and $\bar{x}_i$ is the mean of $x_i(t)$ over time. We selected the E value that gave the highest prediction accuracy (i.e., max $[R^2_E]$ for $E \in \{0, ..., 8\}$) across all values of p.

After we trained the GP-EDM model, we performed predictions for a grid of 200 values of p within $[p_{min}, p_{max}]$ (i.e., interpolated the learned function G_i). To do so for a given value p_j $(j = 1, ..., 200)$, we set initial values of x_i and iterated the learned function G_i forward in time while keeping $p = p_j$ (*SI Appendix*, Fig. S1). Specifically, the prediction $\hat{x}_i(t+1)$ was given by the posterior mean of the GP-EDM model. At every iteration, we used the predicted value as input for the next prediction step. We performed this iteration for 500 time steps and used the last 100 points to characterize the dynamical regime.

We used the JSD to measure the accuracy of our reconstructed bifurcation diagram. The JSD is frequently used to measure the similarity between two probability distributions (e.g., ref. 53). To compute the JSD in a way that is robust to the chosen number of bins, we averaged this metric over several bin numbers (10 to 15 bins). Other procedures such as the Wasserstein distance or the Kolmogorov–Smirnov test could be used here and future work can explore them.

For each value of p_j, we had a sample of two distributions of x_i, one given by the true time series $\{x_i(t)\}$ $(t = 1, ..., T)$ and another given by the predicted time series $\{\hat{x}_i(t)\}$ $(t = 1, ..., T)$, where $T = 100$. Defining P as the discretized (i.e., binned) probability distribution for $\{x_i(t)\}$ and Q as the discretized probability distribution for $\{\hat{x}_i(t)\}$, the JSD is measured as

$$JSD(P||Q) = \frac{1}{2}D(P||M) + \frac{1}{2}D(Q||M), \quad \textbf{[5]}$$

where $M = \frac{1}{2}(P + Q)$ is the mixture distribution of P and Q, and $D(P||M)$ is the Kullback–Leibler divergence of P from M given by

$$D(P||M) = \sum_{x \in \mathcal{X}} P(x) \log_2 \left(\frac{P(x)}{M(x)}\right), \quad \textbf{[6]}$$

where $\mathcal{X}$ is the set of all possible bins and $P(x)$ denotes the probability (i.e., frequency of points) for bin x. Thus, the JSD measures the similarity between P and Q, where 0 denotes very similar distributions and 1 denotes very divergent distributions. Note that the true time series in this case is generated from the model without process noise as opposed to the noisy time series of length $T = 50$ that were used to train the GP-EDM model. That is, we reconstructed the bifurcation diagram from noisy data but compared our reconstruction to the true noise-free diagram.

In addition to the analysis described above, we tested whether our approach can be used as an EWS. To do so, we iteratively applied our approach to a time series generated with the two-species competition model under a slowly changing harvest rate (i.e., the control parameter p). Specifically, we iterated this model while we increased p linearly from 0.42 to 0.48 over 150 time steps. With our parameterization and without process noise, this model shows a fold bifurcation (i.e., a tipping point where the population collapses) at $p = 0.459$.

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Before applying our GP-EDM approach, we performed a change-point analysis (54) to determine the location of the tipping point. We did this because, although the deterministic tipping point occurs at $p = 0.459$, the actual tipping point in the data can occur earlier or later due to process noise. To this end, we split the time series of species i into two windows: $\{x_i(1), ..., x_i(t_{split})\}$ and $\{x_i(t_{split}+1), ..., x_i(T)\}$ for 100 values of t_{split} within [25, 125]. Then, for each t_{split}, we computed the sum of squares of window j (SS_j) using the within-window average, the sum of squares of the entire time series (SS_{all}) using the global average, and the following ratio $SS_{ratio} = \frac{SS_1+SS_2}{SS_{all}}$. We then selected the t_{split} associated with the lowest value of SS_{ratio} as the tipping point. The lower the value of SS_{ratio}, the greater the difference in mean abundance across the two windows.

To apply our GP-EDM approach, we defined 4 time-series windows: 1) $t = 1, ..., 90$, 2) $t = 1, ..., 106$, 3) $t = 1, ..., 110$, and 4) $t = 1, ..., 130$. For each window, we trained our GP-EDM model using information on p up to the end of the window and then performed predictions for a grid of 300 values of p within [0.42, 0.48], as we did in our previous analysis. Note that here, we not only interpolate but also extrapolate G_i beyond observed levels of p. This analysis allowed us to verify how well we can anticipate, not only an upcoming tipping point but also the dynamical regime that lies beyond this point. In addition to our GP-EDM prediction of the tipping point location, we computed a well-studied EWS and compared it to our GP-EDM prediction. We used the variance of $x_i(t)$ for successive rolling windows of length 30 as this EWS, which is expected to increase before a fold bifurcation.

Applying the Approach to an Experimental Microbial Ecosystem. Our first analysis using empirical data consisted of reconstructing bifurcation diagrams of an experimental microbial food web containing one ciliate predator (*Tetrahymena* sp.), a preferred bacterial prey (*Pedobacter* sp.), and a less-preferred bacterial prey (*Brevundimonas* sp.) cultivated in a chemostat (35). This dataset contains time series for all three species cultivated under five distinct dilution rate (i.e., mortality) treatments. Dilution rates of 0.9/day and 0.75/day resulted in stable fixed points, 0.5/day generated chaos, and 0.45/day resulted in a limit cycle (35). We used all experimental replicates, which included one times series for the 0.9/day treatment ($T = 23$), one time series for the 0.75/day treatment ($T = 28$), five time series for the 0.5/day treatment ($T = 30$, $T = 42$, $T = 42$, $T = 47$, and $T = 52$), and two time series for the 0.45/day treatment ($T = 32$ and $T = 45$). Because the only environmental condition changing across treatments is the dilution rate, this dataset represents a family of dynamical systems tuned by a single control parameter, precisely analogous to our simulations shown in Fig. 2.

We applied the same analysis that we performed for model-generated data independently to each species in this system. For a given species, we first concatenated all replicate time series across the four dilution treatments (i.e., $p = 0.45$, 0.5, 0.75, and 0.9) and included dilution rate (p) as an additional input. We removed the 5 initial days of transient dynamics from each replicate to ensure that each time series covered a single dynamical regime. We log-transformed abundance as this greatly improved leave-one-out prediction accuracy. Because dilution rate treatments can result in very different mean abundances, we subtracted the mean treatment abundance from each time series. Then, we standardized all inputs and trained the GP-EDM model. We tested $E \in \{0, ..., 8\}$ and selected the value that maximized leave-one-out prediction accuracy. Note that, because the original number of state variables in this system (e.g., abundances, traits) may be high, large embedding dimensions may be needed to reconstruct the system's attractor via Takens' theorem (51). We then interpolated the GP-EDM model to predict dynamical regimes across a grid of 450 values of p within [0.45, 0.9].

Applying the Approach to a Lake Planktonic Ecosystem. Our second analysis using empirical data consisted of an EWS analysis to anticipate a potential tipping point from an eutrophic to an oligotrophic regime in a lake planktonic food web. We used a dataset of lake planktonic food webs compiled by Merz et al. (16) and selected Lake Zurich to illustrate our approach, as it contained the longest time series and showed signs of a regime shift driven by a gradual decrease in phosphorus. The planktonic food web from Lake Zurich contained monthly time series for 13 functional groups and 2 environmental variables (phosphate concentration and water temperature) from January, 1978 to December, 2019 ($T = 504$). Because our EWS analysis with model-generated data was based on a linearly decreasing control parameter, we applied a centered moving average smoother (55) with a window size of 24 mo to remove seasonality and retain the trend in phosphate concentration. We included seasonality in our GP-EDM model through water temperature (see below). Before applying our GP-EDM approach, we performed a change-point analysis (54) to determine the location of a potential tipping point for each functional group separately (see *Testing approach on model-generated data*). For each functional group, we split its time series into two windows for 300 values of t_{split} within [102, 402] and computed SS_{ratio} for each t_{split}. We selected the t_{split} associated with the lowest value of SS_{ratio} as a potential tipping point.

The above analyses showed that large green algae had the lowest value of SS_{ratio} of all functional groups at $t_{split} = 119$ (November, 1987). Thus, we selected the time-series window from $t = 1$ (January, 1978) to $t = 106$ (October, 1986) to train our GP-EDM model and test whether we could predict the unseen oligotrophic regime. We first performed leave-one-out predictions for each functional group separately using the trained GP-EDM model to verify which group was more predictable. We used up to 7 lags of functional group abundance (i.e., $x_i(t), ..., x_i(t-6)$), up to 2 lags of water temperature (i.e., $w(t), w(t-1)$) to account for seasonality, and 1 lag of phosphate concentration ($p(t)$) (i.e., control parameter) as inputs in our GP-EDM model. We standardized all inputs before training the GP-EDM model. In addition to showing the lowest value of SS_{ratio}, large green algae showed the second-best leave-one-out prediction accuracy ($R^2 = 0.770$). Therefore, we focused on this functional group for subsequent analyses. The best GP-EDM model for large green algae was given by $x_i(t+1) = G_i[x_i(t), ..., x_i(t-3), w(t), w(t-1), p(t)]$.

We then followed our EWS analysis with model-generated data to make predictions for a range of phosphate concentrations using the time series for large green algae. To do so, we defined 4 time-series windows: 1) one year before the potential tipping point ($t = 1, ..., 106$), 2) one month before the potential tipping point ($t = 1, ..., 118$), 3) four years after the potential tipping point ($t = 1, ..., 166$), and 4) nine years after the potential tipping point ($t = 1, ..., 226$). For each window, we trained the GP-EDM model defined above and then performed predictions for a grid of 300 values of p within [0.02, 0.07], which correspond approximately to the minimum and maximum values of phosphate concentration in the data. When iterating predictions forward for a given value of p, we fixed p to that value and set w to the average of the corresponding month. By reconstructing the bifurcation diagram for large green algae as more data becomes available, we provide a real-world illustration of our simulations shown in Fig. 3.

Data, Materials, and Software Availability. Data and code have been deposited in GitHub (https://github.com/lucaspdmedeiros/gpedm-regime-shifts). Previously published data were used for this work (16, 35).

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